

ML & AI NEXUS 2026

1. Round 1 - Kaggle Challenge

Problem Description

Participants must predict the probability of 30-day hospital readmission using a synthetic healthcare dataset. The focus is on building a model that is accurate, well-calibrated, robust, interpretable, and reliable across different groups.

Evaluation Criteria

The main leaderboard metric is **Binary Log Loss**, which rewards accurate probability predictions. Final evaluation will also consider robustness, subgroup performance, statistical validation, explainability, uncertainty, reproducibility, and model limitations. **The goal is to balance predictive performance with trustworthy AI, not just achieve the highest leaderboard score.**

Competition Timeline

Milestone	Date & Time
Start Date	Sunday, 13th September 2026 - 8:00 AM (competition link sent via email)
End Date	Monday, 14th September 2026 - 8:00 AM (24 hours from start)
Submission Limit	Maximum 10 submissions per day while the competition window is open

Competition Link will be shared Through an email and also through the WhatsApp group at 8.00 am on 13th September, Sunday

Rules of the Competition

- Only one Kaggle account is to be created per team. Teams cannot sign up to Kaggle from multiple accounts and therefore cannot submit from multiple accounts.
- Use your TEAM NAME as the Kaggle SCREEN NAME / USERNAME so the organizers can identify you on the leaderboard correctly.
- After logging into Kaggle, click on the given link to go to the competition page and start working on it .
- There will be a maximum of **10 submissions per team**.
 - **First 5 submissions:** From Sunday 8.00 a.m. to Sunday midnight
 - **Next 5 submissions:** From Sunday midnight to Monday 8.00 a.m.
- So, each team can make **up to 10 submissions in total**.
- No private sharing of the response variable of the test dataset is allowed. This will lead to disqualification.
- Team mergers are not allowed in this competition.

- All participants must follow standard Kaggle guidelines and the competition instructions.
- Private sharing of code or external datasets outside the registered team is strictly prohibited.
- Teams must follow the announced submission deadline, format, and upload method.

Description of Attached Files

A training set and a test set will be provided along with the data description

Kaggle Leaderboard Mark Allocation (40 Marks)

For the **Kaggle Competition (40 marks)**, the evaluation will be divided into two components:

- **25 marks** - Kaggle Leaderboard Score
- **15 marks** - Submitted **Trust Card**

The **Trust Card** and your team's notebook submission link: <https://forms.gle/6rcYfFzsz5L94Pwn7>

Submission Deadline: 14 September 2026 (Wednesday), 8:00 AM

The **Trust Card template** is available in the competition link. Teams are expected to follow the given template when preparing their submission.

Purpose of the Trust Card :

This card summarizes why a predictive model should or should not be trusted. It should complement the leaderboard score by documenting validation, calibration, robustness, uncertainty, explainability, and limitations.

1. Model Summary
2. Validation Strategy
3. Predictive Performance
4. Calibration
5. Robustness and Stability
6. Subgroup Reliability
7. Uncertainty and Human Oversight
8. Explainability
9. Model Comparison
10. Failure Modes and Limitations
11. Deployment / Use Recommendation
12. Reproducibility

Leaderboard Rank	Marks (out of 25)
1	25.0
2	24.0
3	23.0
4	22.0
5	21.0
6	20.0
7	19.0
8	18.0
9	17.0
10	16.0
11	15.0
12	14.0
13	13.0
14	12.0
15	11.0
16	10.0
17	9.0
18	8.0

- Ties on the leaderboard (identical scores) receive identical marks, corresponding to the higher rank position.
- Teams that do not submit a valid entry to the leaderboard receive 0 marks for this component and are not eligible for the presentation/viva stage.
- Rest of the **marks (15 marks out of 40) will be awarded based on your team's submitted Trust Card.**

Presentation & Viva Session

Eligibility Criteria

- Only teams that (a) complete the Kaggle Challenge and (b) upload their final presentation on time will be considered for the viva session.
- Teams must communicate the important technical and practical evidence through the presentation and viva.
- **Submission deadline for the presentation:** Wednesday, 16th September 2026, by 11:59 PM.
- **Submission Link:** <https://forms.gle/ZBa72adq7aHei7iu9>

- The time allocation (slot) for each team's presentation and viva will be shared separately after the submission deadline.

Presentation Format

Total duration: 15 minutes per team - 10 minutes for slides, 5 minutes for the viva.

The presentation should briefly include:

1. Team name, university or institute.
2. An introduction to the problem and the related industry challenge in the team's point of view.
3. A short description of the dataset and the main exploratory findings.
4. The analytical approach, preprocessing, model architecture, validation method, and reasons for selecting the final model.
5. The main results and Kaggle performance, compared with a suitable baseline where relevant.
6. Problems encountered during the analysis, the solutions used, and suggestions for overcoming the remaining problems.
7. The practical value of the solution and a short conclusion.

Trustworthy and Interpretable Models

Please keep your presentation brief and focus mainly on the **interpretation of your results**. Avoid spending too much time explaining the methodology or technical details.

Overall Evaluation Structure

Assessment Component	Weight	Main Focus
Kaggle Leaderboard Challenge	40%	Predictive performance
Innovative Pitch Presentation	40%	Approach, analysis, innovation and communication
Viva Session	20%	Understanding and ability to defend the work

The three scores are combined to determine the final standing. A high Kaggle score alone does not determine the final standing.

Presentation Mark Allocation (40 Marks)

Weighted toward this year's theme -Trustworthy AI, causal inference, statistical reasoning, and explainable AI - alongside sound analytical practice and clear communication.

Criterion	Marks
Problem understanding & industry perspective	4
Dataset description & exploratory findings	4
Analytical approach: preprocessing, model architecture & validation method	8

Results & Kaggle performance, compared with a suitable baseline	6
Model trustworthiness, interpretability, fairness, uncertainty & limitations	8
Challenges faced, solutions used & suggestions for improvement	4
Practical value, innovation & conclusion	4
Clarity of communication & time management (10 min)	2
Total	40

Viva Mark Allocation (20 Marks)

Criterion	Marks
Depth of understanding of the team's own work and analysis	6
Ability to justify and defend methodology, model, and design choices	6
Quality of responses on trustworthiness, interpretability, and the Stat Day theme	5
Team coordination and clarity in answering under time pressure	3
Total	20

Final Score Combination

Final Score (out of 100) = Kaggle Leaderboard Marks (40) + Presentation Marks (40) + Viva Marks (20).

- The three component scores are added directly (already expressed out of 40, 40, and 20) to produce the Final Score out of 100.
- Final standings are ranked by this combined Final Score. A high Kaggle score alone does not determine the final standing - strong performance across all three components is required to place highly.